3.2 Displacement and Velocity

Chapter 3 – Horizontal Linear Motion: worked answers

Question 1

Speed = distance ÷ time = 200 ÷ 5

Answer: 40 m s^{-1}

Question 2

Time = 2 × 60 = 120 s

Displacement = $v \times t$ = 3 × 120

Answer: 360 m east

Question 3

Displacement = $v \times t$, with v = 6 m s^{-1} west

(i) 6 × 1 = **6 m west**

(ii) 6 × 60 = **360 m west**

(iii) 6 × 3600 = **21 600 m west** (21.6 km)

Question 4

The wheel is 20 m tall, so its diameter is 20 m.

(i) Three full cycles return the kid to the starting point.

Total displacement = 0 m

(ii) One cycle = circumference = $\pi \times 20$ = 62.8 m

Three cycles = 3 × 62.8 = **188.5 m** (about 189 m)

(iii) Speed = distance ÷ time = 188.5 ÷ 60 = **3.14 m s^{-1}**

Question 5

The displacements are at right angles, so use Pythagoras:

Total displacement = $\sqrt{(5^2 + 12^2)} = \sqrt{169}$ = **13 m**

Direction: tan θ = 12 ÷ 5, so θ = 67.4°, i.e. **67.4° north of east**

Average velocity = displacement ÷ time = 13 ÷ 10 = **1.3 m s^{-1}** at 67.4° north of east

Note: distance is a scalar (Q4 (ii): 188.5 m), whereas displacement is a vector (Q4 (i): zero).